L02: Basic Estimators and Experimental Pitfalls

S&DS 6165: Topics in Causal Inference

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Lecture brief: View the source `brief.md` · How these notes are made

Lecture outline

- Use a four-unit experiment to separate an estimator's randomization variance from an estimated standard error.
- Introduce inverse-probability weighting and see why random total weight can make an unbiased estimator unstable.
- Compare the HT and Hájek variances, then develop adaptive normalization as a more flexible response to random total weight.
- Reinterpret regression adjustment, CUPED, GREG, and calibration as related control-variate strategies.

1 From an unbiased estimator to an uncertainty statement

Under complete randomization with fixed arm sizes N_1 and N_0, L01 established that

$$\hat{\tau}_{\mathrm{DM}} = \frac{1}{N_1}\sum_{i=1}^{N} Z_i Y_i^{\mathrm{obs}} - \frac{1}{N_0}\sum_{i=1}^{N}(1-Z_i)Y_i^{\mathrm{obs}}$$

is design-unbiased for the finite-population average treatment effect τ. Unbiasedness is a statement about the center of the estimator's randomization distribution. It does not say how dispersed that distribution is.

For fixed potential outcomes, define the finite-population variances

$$S_z^2 = \frac{1}{N-1}\sum_{i=1}^{N}\{Y_i(z) - \overline{Y}(z)\}^2, \qquad S_\tau^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\tau_i - \tau)^2,$$

where $\tau_i = Y_i(1) - Y_i(0)$. These are fixed features of the science table, not variances over a model for the outcomes. Neyman's exact randomization variance is (Splawa-Neyman et al. 1990)

$$V_Z := \mathrm{Var}_Z(\hat{\tau}_{\mathrm{DM}}) = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} - \frac{S_\tau^2}{N}. \tag{1}$$

After assignment, the observed sample variance in arm z, when $N_z > 1$, is

$$s_z^2 = \frac{1}{N_z - 1}\sum_{i:Z_i=z}\{Y_i^{\mathrm{obs}} - \overline{Y}_z^{\mathrm{obs}}\}^2.$$

Keep three objects separate:

1. V_Z is a fixed but generally unknown property of the design and science table.
2. $\widehat{V}_N = s_1^2/N_1 + s_0^2/N_0$ is a random **variance estimator**, where s_z^2 is the sample variance of the observed outcomes in arm z.

3. After assignment, $\widehat{V}_N^{\text{obs}}$ is the realized **variance estimate**, and $\sqrt{\widehat{V}_N^{\text{obs}}}$ is the reported standard error.

Because the individual effects are not jointly observed, S_τ^2 is generally unidentified. The Neyman estimator drops its negative contribution. Under complete randomization,

$$\mathbb{E}_Z(\widehat{V}_N) - V_Z = \frac{S_\tau^2}{N} \geq 0,$$

so it is conservative. When treatment effects are constant, $S_\tau^2 = 0$ and $\widehat{V}_N$ is unbiased for V_Z; this does not mean that every realized variance estimate equals the true randomization variance.

2 A four-unit experiment by hand

Consider the fixed science table

Unit	$Y_i(0)$	$Y_i(1)$	τ_i
1	1	3	2
2	2	3	1
3	4	5	1
4	5	9	4

The finite-population ATE is $\tau = 2$. Treat exactly two units. The six equally likely assignments give

Treated units	$\widehat{\tau}_{\text{DM}}$
$\{1,2\}$	-1.5
$\{1,3\}$	0.5
$\{1,4\}$	3.0
$\{2,3\}$	1.0
$\{2,4\}$	3.5
$\{3,4\}$	5.5

For example, if units 1 and 4 are treated, their observed treated outcomes are 3 and 9, while the observed control outcomes for units 2 and 3 are 2 and 4. The treated and control means are therefore 6 and 3, giving $\widehat{\tau}_{\text{DM}} = 3$. The calculation for every row uses the same fixed science table but reveals a different pair of entries from each potential-outcome column.

The six estimates average to 2. Their variance, using the randomization probabilities rather than a sample-variance denominator, is

$$\frac{1}{6}\sum_{\mathbf{z}\in\mathcal{Z}}\{\widehat{\tau}_{\text{DM}}(\mathbf{z}) - 2\}^2 = \frac{31}{6}.$$

The same answer follows from Equation 1: $S_1^2 = 8$, $S_0^2 = 10/3$, and $S_\tau^2 = 2$. If units 1 and 4 are treated, the realized within-arm variances are $s_1^2 = 18$ and $s_0^2 = 2$, so $\widehat{V}_N^{\text{obs}} = 10$ and the reported standard error is $\sqrt{10}$. The true randomization variance remains $31/6$; one realized variance estimate need not equal it.

This tiny example is worth reconstructing at the board because it makes the source of every probability statement visible. There is one science table, six possible assignments, and a different observed estimate under each assignment.

3 Inverse-probability weighting

Complete randomization gave every unit the same treatment probability. More general designs need not. An experiment might deliberately assign a scarce treatment less often to expensive units, or use strata with different treatment fractions. The units whose potential outcomes are revealed under a given treatment state would then represent unequal amounts of the full experimental population. Inverse-probability weighting corrects that unequal representation.

Let

$$\pi_i = \mathbb{P}_Z(Z_i = 1), \qquad 0 < \pi_i < 1.$$

The **Horvitz–Thompson** (HT) estimators of the two finite-population means are

$$\widehat{\mu}_{1,\text{HT}} = \frac{1}{N}\sum_{i=1}^{N} \frac{Z_i Y_i^{\text{obs}}}{\pi_i}, \qquad \widehat{\mu}_{0,\text{HT}} = \frac{1}{N}\sum_{i=1}^{N} \frac{(1-Z_i) Y_i^{\text{obs}}}{1-\pi_i},$$

and $\widehat{\tau}_{\text{HT}} = \widehat{\mu}_{1,\text{HT}} - \widehat{\mu}_{0,\text{HT}}$ (Horvitz and Thompson 1952). A treated unit with $\pi_i = 0.2$, for example, receives weight $1/0.2 = 5$: it is observed under treatment only one fifth of the time across repeated assignments, so each such observation represents five assignment opportunities. Formally, each unit's weighted treatment indicator has expectation one,

$$\mathbb{E}_Z\left(\frac{Z_i}{\pi_i}\right) = 1.$$

Linearity of expectation then gives

$$\mathbb{E}_Z(\widehat{\mu}_{1,\text{HT}}) = \frac{1}{N}\sum_i Y_i(1) = \overline{Y}(1),$$

and similarly for control. This argument uses only the marginal inclusion probabilities; the Z_i need not be independent. Positivity is essential, because a potential outcome assigned probability zero can never receive an inverse-probability correction. The conclusion is **unconditional** design unbiasedness: it averages over every assignment in the design, including the all-treated and all-control assignments that a Bernoulli design permits.

Under complete randomization, $\pi_i = N_1/N$, so

$$\hat{\mu}_{1,\mathrm{HT}} = \frac{1}{N_1} \sum_i Z_i Y_i^{\mathrm{obs}}.$$

Thus HT and difference in means coincide. Under i.i.d. Bernoulli assignment, they do not.

3.1 The Bernoulli subtlety

With common Bernoulli probability p, the usual difference in means divides by the random counts N_1 and N_0. It is undefined when either count is zero. Conditional on any $N_1 = n_1 \in \{1, \dots, N-1\}$, however, the assignment is a complete randomization with n_1 treated units. Hence

$$\mathbb{E}_Z(\hat{\tau}_{\mathrm{DM}} \mid N_1 = n_1) = \tau.$$

It is therefore conditionally unbiased on every nondegenerate arm size, but it is not an everywhere-defined, unconditionally unbiased estimator over the original Bernoulli design. By contrast,

$$\hat{\tau}_{\mathrm{HT}} = \frac{1}{N} \sum_i \left\{ \frac{Z_i Y_i^{\mathrm{obs}}}{p} - \frac{(1 - Z_i) Y_i^{\mathrm{obs}}}{1 - p} \right\}$$

is defined and design-unbiased on the full assignment space.

> **i** What HT is correcting
>
> Inverse-probability weighting gives a unit that is rarely assigned to a particular treatment state more representation when that state occurs. The identity $\mathbb{E}_Z(Z_i/\pi_i) = 1$ is the entire unbiasedness argument for the treated potential-outcome mean. The cost is that small probabilities create large and noisy weights.

4 Hájek normalization

HT weights have the right total **in expectation**, not in every assignment. For the treated mean, their realized sum is

$$\widehat{N}_1^w = \sum_i \frac{Z_i}{\pi_i}, \qquad \mathbb{E}_Z(\widehat{N}_1^w) = N.$$

The notation $\widehat{N}_1^w$ does not denote the number of treated units. It is the population size represented by the treated units after weighting. HT divides its weighted outcome total by the known N even when $\widehat{N}_1^w \neq N$. The Hájek estimator instead divides by the realized total weight (Hájek 1971):

$$\hat{\mu}_{1,\mathrm{H}} = \frac{\sum_i Z_i Y_i^{\mathrm{obs}}/\pi_i}{\sum_i Z_i/\pi_i}, \tag{2}$$

with an analogous control estimator. On an empty arm its denominator is zero. Under common-probability Bernoulli assignment, the Hájek arm means are the ordinary arm means. Under complete randomization, HT, Hájek, and difference in means all coincide.

With unequal probabilities, normalization can alter both finite-sample bias and variance; neither change has a universal sign. We compare assignment variances first, then turn to ratio bias from the random denominator.

4.1 A first variance comparison: HT versus Hájek

To see what normalization changes without carrying treatment and control notation through every line, isolate one generic finite-population mean under independent Bernoulli sampling. Let $I_i \sim \text{Bernoulli}(\pi_i)$ independently reveal fixed values Y_i. The target total is $T_Y = \sum_i Y_i = N\mu$. Its inverse-probability estimate and the corresponding estimated population size are

$$\hat{T}_Y = \sum_i \frac{I_i Y_i}{\pi_i}, \qquad \widehat{N} = \sum_i \frac{I_i}{\pi_i}, \qquad \mu = \frac{1}{N}\sum_i Y_i.$$

This is the treated-arm problem with $I_i = Z_i$ and $Y_i = Y_i(1)$. For the control arm, take $I_i = 1 - Z_i$, $Y_i = Y_i(0)$, and replace π_i by $1 - \pi_i$. HT is $\hat{T}_Y/N$, whereas Hájek is $\hat{T}_Y/\widehat{N}$. Both numerators estimate the same total; only the denominator differs.

For HT, the estimation error has the exact representation

$$\hat{\mu}_{\text{HT}} - \mu = \frac{1}{N}\sum_i \left(\frac{I_i}{\pi_i} - 1\right) Y_i.$$

Independence makes the cross-unit covariance terms vanish, giving the exact variance

$$\text{Var}_Z(\hat{\mu}_{\text{HT}}) = \frac{1}{N^2}\sum_i \frac{1-\pi_i}{\pi_i} Y_i^2. \tag{3}$$

Writing $Y_i = (Y_i - \mu) + \mu$ separates two sources of error:

$$\hat{\mu}_{\text{HT}} - \mu = \frac{1}{N}\sum_i \left(\frac{I_i}{\pi_i} - 1\right)(Y_i - \mu) + \mu\left(\frac{\widehat{N}}{N} - 1\right). \tag{4}$$

The second term is produced by the random total weight. If the same constant a were added to every Y_i, the target mean would increase by a, but the HT estimation error would also acquire $a(\widehat{N}/N - 1)$. Thus HT variance can depend on the common additive baseline of the outcomes, not only on how dispersed they are around their mean.

When $\widehat{N} > 0$, Hájek divides by the realized total weight, so the same additive shift simply adds a to both the estimate and its target. Its estimation error is unchanged. The price is a random denominator, which makes an exact finite-sample variance less transparent.

The Taylor calculation shows how normalization removes the random-baseline term. Apply $f(t, n) = t/n$ to $(\hat{T}_Y, \widehat{N})$ and expand around the expectations $(T_Y, N) = (N\mu, N)$. Because the two derivatives at that point are $1/N$ and $-\mu/N$,

$$\frac{\hat{T}_Y}{\widehat{N}} \approx \mu + \frac{\hat{T}_Y - T_Y}{N} - \frac{\mu}{N}(\widehat{N} - N).$$

Substituting the two weighted totals and collecting each unit's contribution gives

$$\hat{\mu}_{\mathrm{H}} - \mu \approx \frac{1}{N} \sum_i \left(\frac{I_i}{\pi_i} - 1 \right) (Y_i - \mu). \tag{5}$$

Consequently, its first-order variance is

$$\mathrm{Var}_Z(\hat{\mu}_{\mathrm{H}}) \approx \frac{1}{N^2} \sum_i \frac{1 - \pi_i}{\pi_i} (Y_i - \mu)^2. \tag{6}$$

The contrast between Equation 3 and Equation 6 is now explicit: HT's exact variance involves the squared outcomes Y_i^2, whereas the first-order Hájek variance involves squared deviations $(Y_i - \mu)^2$. With common $\pi_i = p$,

$$\mathrm{Var}_Z(\hat{\mu}_{\mathrm{HT}}) - \mathrm{Var}_Z(\hat{\mu}_{\mathrm{H}}) \approx \frac{1 - p}{pN} \mu^2.$$

The difference is large when the outcome mean is far from zero relative to the outcomes' dispersion. Equivalently, a common additive shift changes HT's variance because random total weight multiplies that shift, while it leaves the first-order Hájek variance unchanged. This is the classic survey-sampling linearization logic (Särndal et al. 1989, 1992). The calculation isolates one arm; for the treatment-effect estimator, the treated- and control-arm variances must also be combined with their design covariance.

4.2 Ratio bias from normalization

Variance is not the entire comparison. A second-order expansion gives the approximate ratio bias

$$\mathrm{Bias}_Z(\hat{\mu}_{\mathrm{H}}) \approx \frac{\mu \, \mathrm{Var}_Z(\widehat{N}) - \mathrm{Cov}_Z(\hat{T}_Y, \widehat{N})}{N^2}.$$

Here

$$\mathrm{Var}_Z(\widehat{N}) = \sum_i \frac{1 - \pi_i}{\pi_i}, \qquad \mathrm{Cov}_Z(\hat{T}_Y, \widehat{N}) = \sum_i \frac{1 - \pi_i}{\pi_i} Y_i.$$

The expression is informative even before it is evaluated: ratio bias appears when fluctuations in the estimated outcome total and the estimated population size do not line up as μ times one another. For common $\pi_i = p$, the two terms cancel. This agrees with the exact fact that, conditional on any

positive realized sample size, the Hájek mean is the ordinary sample mean under simple random sampling. The Taylor calculations nevertheless remain approximations to an unconditional ratio problem whose denominator can be zero. They are most informative when $\widehat{N}$ is concentrated near N. With small effective samples, extreme probabilities, or influential units, higher-order terms can matter and neither estimator has a universal MSE advantage.

4.3 HT versus Hájek in a finite population

The demonstration uses

$$Y_i(0) = b + \sigma x_i, \qquad Y_i(1) = b + \sigma x_i + \tau,$$

with finite-population x_i centered at zero. The common baseline b shifts every potential outcome by the same amount, so it changes neither the individual treatment effects nor the ATE. For an HT arm mean, however, that shift adds the random error term $b(\widehat{N}/N - 1)$. Because Hájek divides by $\widehat{N}$, the same shift simply adds b to its arm-mean estimate and does not change that estimate's variance. When $b = 0$, this first-order shift-cancellation advantage disappears; the random Hájek denominator can then produce more conditional MSE than HT's fixed denominator.

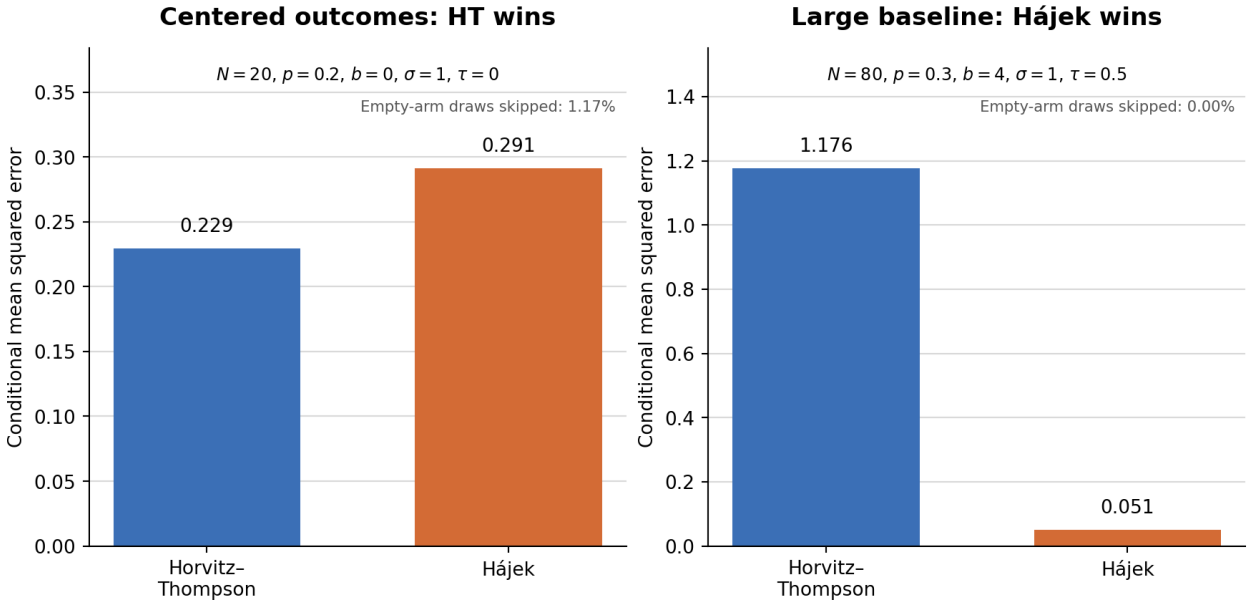


Figure 1: Two fixed simulation settings show that neither estimator uniformly minimizes finite-sample MSE. HT wins in the centered setting; Hájek wins when outcomes have a large common baseline.

The left panel uses centered outcomes with no effect; the right introduces a large common baseline. The reversal isolates the random-total-weight mechanism in Equation 4.

5 Normalization is a continuum

HT leaves the realized total-weight error uncorrected; Hájek removes it completely. That contrast suggests a less binary question: how much of the denominator fluctuation should an estimator use? Trotter and Tukey introduced an affine family that makes this amount explicit; Khan and Ugander (2023) develop the family for inverse-probability weighting. For the same generic mean problem, define

$$\hat{\mu}_\lambda = \frac{\hat{T}_Y}{(1-\lambda)N + \lambda\widehat{N}} = \frac{\hat{T}_Y}{N + \lambda(\widehat{N} - N)}. \tag{7}$$

$\lambda = 0$ gives HT and $\lambda = 1$ gives Hájek. The second expression makes the role of λ visible: it controls how strongly the random error $\widehat{N} - N$ enters the denominator. Values outside $[0, 1]$ are allowed; the family is an affine normalization family, not a convex interpolation. In a realized finite sample, however, a chosen value is usable only when $N + \lambda(\widehat{N} - N) \neq 0$. Extrapolated values can drive this denominator near zero or make it negative, so a first-order variance optimum is not by itself a finite-sample stability guarantee.

The same first-order expansion used for Hájek yields

$$\hat{\mu}_\lambda - \mu \approx \frac{1}{N}\sum_i \left(\frac{I_i}{\pi_i} - 1\right)(Y_i - \lambda\mu). \tag{8}$$

This equation turns the choice of denominator into the choice of a centering constant. HT uses no centering, because $\lambda\mu = 0$. Hájek centers at the target mean, because $\lambda\mu = \mu$. An intermediate or extrapolated value uses a different constant.

Before changing probability frameworks, the optimal first-order choice can be written entirely in finite-population notation. Let $q_i = (1 - \pi_i)/\pi_i = \mathrm{Var}_Z(I_i/\pi_i)$. Under independent Bernoulli inclusion, Equation 8 gives

$$\mathrm{Var}_Z(\hat{\mu}_\lambda) \approx \frac{1}{N^2}\sum_i q_i(Y_i - \lambda\mu)^2.$$

When $\mu \neq 0$ and at least one $q_i > 0$, minimizing this quadratic yields

$$\lambda_N^* = \frac{\sum_i q_i Y_i}{\mu \sum_i q_i}, \qquad c_N^* := \lambda_N^*\mu = \frac{\sum_i q_i Y_i}{\sum_i q_i}. \tag{9}$$

Thus the useful object is not fundamentally a fraction λ. It is the centering constant c_N^*, a weighted mean of the fixed Y_i that emphasizes units with noisier inverse-probability weights.

To state the adaptive choice used by Khan and Ugander (2023), we now make an explicit change of framework. The preceding variance calculations held a finite population fixed. Their asymptotic analysis instead treats the pairs (Y_i, π_i) as i.i.d. draws of a generic pair (Y, P), where P is a random inclusion probability. In this paragraph, $\mu = \mathbb{E}(Y)$ is the superpopulation mean rather than the mean of one fixed finite population. Define the weight-noise factor $Q = (1 - P)/P$ and the two population moments

$$\Pi = \mathbb{E}(Q), \qquad A = \mathbb{E}(YQ).$$

These expectations are the superpopulation analogues of $N^{-1}\sum_i q_i$ and $N^{-1}\sum_i q_i Y_i$ in Equation 9.

For the usual $\sqrt{N}$ scaling, the part of the first-order asymptotic variance that depends on λ is

$$\sigma_\lambda^2 = \mathbb{E}\{Q(Y - \lambda\mu)^2\}. \tag{10}$$

Superpopulation sampling also contributes variation in Y itself, but that term does not depend on λ and therefore does not affect which member of the normalization family minimizes variance.

This is a quadratic in λ. When $\mu \neq 0$ and $\Pi > 0$, differentiating it gives

$$\frac{d\sigma_\lambda^2}{d\lambda} = -2\mu(A - \lambda\mu\Pi),$$

and hence the minimizer

$$\lambda^* = \frac{A}{\Pi\mu}. \tag{11}$$

The numerator and denominator become easier to interpret after absorbing the factor μ. Define the optimal centering constant

$$c^* = \frac{A}{\Pi}$$

so that $\lambda^* = c^*/\mu$. The constant c^* is an average of Y under weights proportional to Q: it emphasizes outcomes attached to the inclusion probabilities that create the most inverse-weight noise. If the inclusion probability is common, Q is constant, $c^* = \mu$, and Hájek is optimal within this first-order family. With heterogeneous inclusion probabilities, the high-weight portion of the population can have a mean different from μ; then neither HT nor Hájek need use the best centering constant, and λ^* need not lie between 0 and 1.

When $\mu = 0$, the product $\lambda\mu$ vanishes in Equation 8, so the first-order variance does not depend on λ. This does not mean that normalization is irrelevant in every finite sample. It means that λ, which divides a useful centering constant by μ, is a poor coordinate near zero. The constant c^* remains well defined.

The population moments A and Π are unknown in the generic inclusion problem. The identity $\mathbb{E}(I_i Y_i/\pi_i \mid Y_i, \pi_i) = Y_i$ motivates the IPW sample analogues

$$\widehat{A} = \frac{1}{N}\sum_i \frac{1-\pi_i}{\pi_i}\frac{I_i Y_i}{\pi_i}, \qquad \widehat{\Pi} = \frac{1}{N}\sum_i \frac{1-\pi_i}{\pi_i}\frac{I_i}{\pi_i}, \qquad \hat{c} = \frac{\widehat{A}}{\widehat{\Pi}}.$$

The remaining obstacle is μ in $\lambda^* = c^*/\mu$. Adaptive normalization uses the current mean estimate in that denominator, updates the normalization, and repeats. Starting, for example, at $\hat{\mu}^{(0)} = \hat{\mu}_{\mathrm{HT}}$, the iteration sets

$$\hat{\lambda}^{(t)} = \frac{\hat{c}}{\hat{\mu}^{(t-1)}}, \qquad \hat{\mu}^{(t)} = \frac{\hat{T}_Y}{(1-\hat{\lambda}^{(t)})N + \hat{\lambda}^{(t)}\widehat{N}}.$$

The two equations form a feedback loop: the current mean estimate supplies the denominator of $\hat{\lambda}^{(t)}$, and the resulting normalization supplies the next mean estimate. At a nonzero fixed point, substitute $\hat{\lambda} = \hat{c}/\hat{\mu}$ into the denominator and multiply through:

$$N\hat{\mu} + \hat{c}(\widehat{N} - N) = \hat{T}_Y.$$

Solving for $\hat{\mu}$ gives the closed form

$$\hat{\mu}_{\mathrm{AN}} = \frac{\hat{T}_Y}{N} + \hat{c}\left(1 - \frac{\widehat{N}}{N}\right) = \hat{\mu}_{\mathrm{HT}} - \hat{c}\left(\frac{\widehat{N}}{N} - 1\right). \tag{12}$$

The fixed point is therefore a **control-variate**, or regression-control, estimator. The quantity $D = \widehat{N}/N - 1$ has design expectation zero and is correlated with the part of HT error generated by total-weight fluctuation. For any *fixed* coefficient c, subtracting cD preserves exact design unbiasedness; choosing the population coefficient c^* minimizes the first-order variance in this one-dimensional control family. HT uses $c = 0$, Hájek corresponds at first order to $c = \mu$, and adaptive normalization plugs in $\hat{c}$.

Because $\hat{c}$ is estimated from the same realization as D, the mean-zero property of D alone does not prove exact unbiasedness of the adaptive estimator. The guarantee in Khan and Ugander (2023) is asymptotic: under their moment and consistency conditions, the adaptive variance is no larger than the corresponding HT or Hájek variance. Unstable estimated moments or a nearly empty effective sample can still matter in finite samples. The closed form in Equation 12 is also preferable if an iterate passes near zero, where the λ parameterization becomes unstable.

6 Regression adjustment

The weighted-population-size error is one random quantity whose expectation is known from the design. Pretreatment covariates provide many more. Suppose, for example, that the treated arm happens to contain users who were more active before an experiment. That difference cannot have been caused by treatment, but if prior activity predicts the outcome, it can make the unadjusted outcome difference unusually large or small.

Under complete randomization with fixed positive arm sizes, define the realized covariate means

$$\overline{X}_1^{\mathrm{obs}} = \frac{1}{N_1}\sum_i Z_i X_i, \qquad \overline{X}_0^{\mathrm{obs}} = \frac{1}{N_0}\sum_i (1 - Z_i) X_i.$$

The covariates are vectors, so these are componentwise arm averages. Their difference is observable after assignment. The superscript obs refers to the realized treatment and control groups; it does not suggest that X_i was measured after treatment. We write the difference as

$$\hat{\tau}_X = \overline{X}_1^{\text{obs}} - \overline{X}_0^{\text{obs}}$$

This difference has design expectation zero: randomization does not guarantee balance in the realized experiment, but it makes the imbalance average to zero across assignments. If β maps a one-unit covariate imbalance into predicted outcome imbalance, the natural correction is

$$\hat{\tau}_{\text{adj}}(\beta) = \hat{\tau}_{\text{DM}} - \beta^\top \hat{\tau}_X. \tag{13}$$

For any coefficient β fixed before assignment, $\mathbb{E}_Z\{\hat{\tau}_{\text{adj}}(\beta)\} = \tau$ follows immediately: the adjustment subtracts a fixed multiple of a mean-zero random quantity. When X predicts outcomes, this can remove chance imbalance that would otherwise masquerade as treatment effect. An arbitrary coefficient can instead add noise; design unbiasedness and variance reduction are separate claims.

A common slope is not required. If $\overline{X} = N^{-1}\sum_i X_i$ is known, fixed treatment- and control-arm coefficients give the following estimator, where $\overline{Y}_z^{\text{obs}}$ denotes the observed outcome mean in arm z:

$$\begin{aligned}\hat{\tau}_{\text{adj}}(\beta_1, \beta_0) = &\left\{\overline{Y}_1^{\text{obs}} - (\overline{X}_1^{\text{obs}} - \overline{X})^\top \beta_1\right\} \\ &- \left\{\overline{Y}_0^{\text{obs}} - (\overline{X}_0^{\text{obs}} - \overline{X})^\top \beta_0\right\}.\end{aligned} \tag{14}$$

For fixed coefficients, each bracket is design-unbiased for its corresponding finite-population potential-outcome mean. Taking $\beta_1 = \beta_0 = \beta$ recovers Equation 13. In practice, the coefficients are usually fitted from the experimental outcomes. They are then assignment-dependent, so the exact finite-sample mean-zero argument no longer follows merely from $\mathbb{E}_Z(\hat{\tau}_X) = 0$. The regression should allow treatment-specific slopes—equivalently, include treatment-by-centered-covariate interactions—because the two potential-outcome surfaces need not be parallel (Freedman 2008; Lin 2013). Randomization protects the large-sample causal interpretation under suitable regularity conditions, but careless specifications or standard errors can lose efficiency or misstate uncertainty.

This returns us to normalization. In Equation 12, $\widehat{N}/N - 1$ is the inverse-weighted imbalance in the constant feature 1: its target mean is exactly one, but its realized weighted mean need not be. Adaptive normalization uses that scalar imbalance as a control variable. Conventional regression adjustment applies the same logic to informative pretreatment features. GREG makes the connection especially explicit.

6.1 GREG from a known baseline total

Suppose an online platform fixes a population of N accounts before an experiment. For every account it already knows X_i, the number of active days during the preceding four weeks. The finite-population mean

$$\overline{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$$

is therefore known exactly before assignment. By contrast, $Y_i(1)$, the experiment-period outcome under treatment, is observed only for treated accounts.

The treated accounts may happen to include unusually active users. Under complete randomization, for example, their pre-period mean might be 4.8 active days even though the full roster mean is 4.0. Randomization makes this discrepancy zero on average across assignments, but it is present in the experiment that actually occurred. If a prespecified relationship says that one additional pre-period active day predicts roughly two units of the experiment-period outcome, the 0.8-day imbalance suggests about 1.6 units of chance inflation in the treated outcome mean.

The same idea extends to a vector of pretreatment covariates. Define its HT estimate within the treated arm,

$$\widehat{\overline{X}}_{1,\mathrm{HT}} = \frac{1}{N}\sum_i \frac{Z_i X_i}{\pi_i}, \qquad \mathbb{E}_Z\Big(\widehat{\overline{X}}_{1,\mathrm{HT}}\Big) = \overline{X}.$$

The discrepancy $\overline{X} - \widehat{\overline{X}}_{1,\mathrm{HT}}$ is observable and has design expectation zero. If β converts covariate imbalance into predicted outcome imbalance, the natural correction is

$$\hat{\mu}_{1,\mathrm{GREG}}(\beta) = \hat{\mu}_{1,\mathrm{HT}} + (\overline{X} - \widehat{\overline{X}}_{1,\mathrm{HT}})^\top \beta. \tag{15}$$

Under complete randomization, $\widehat{\overline{X}}_{1,\mathrm{HT}}$ is simply the ordinary treated-arm covariate mean, so it is the 4.8 in the example. For a coefficient β fixed relative to assignment, Equation 15 is the **generalized-difference**, or regression-control, construction at the heart of GREG. Replacing β by a weighted regression estimate gives the usual **generalized regression estimator** (GREG) (Cassel et al. 1976). In the scalar example, $\beta = 2$ makes the correction $(4.0 - 4.8)2 = -1.6$.

Writing $m_1(x) = x^\top \beta$ gives an equivalent prediction-plus-residual form:

$$\hat{\mu}_{1,\mathrm{GREG}}(m_1) = \frac{1}{N}\sum_i m_1(X_i) + \frac{1}{N}\sum_i \frac{Z_i}{\pi_i}\{Y_i^{\mathrm{obs}} - m_1(X_i)\}. \tag{16}$$

The two terms divide the work. The full roster supplies the average prediction, which can be computed without observing any experiment-period outcomes. Inverse-probability weighting is then applied only to the prediction errors among treated units. If an intercept is desired, augment X_i with a leading 1. Under unequal probabilities, correcting imbalance in that intercept is exactly a correction for $\widehat{N}/N - 1$; under complete randomization, the intercept is already balanced. For β fixed before assignment,

$$\hat{\mu}_{1,\mathrm{GREG}}(\beta) - \overline{Y}(1) = \frac{1}{N}\sum_i \left(\frac{Z_i}{\pi_i} - 1\right)\{Y_i(1) - X_i^\top \beta\}.$$

Exact design unbiasedness therefore does not require the linear prediction to be correct: misspecification changes the residuals, but the IPW correction still centers their contribution correctly. Under independent Bernoulli assignment, the exact variance is

$$\mathrm{Var}_Z\{\hat{\mu}_{1,\mathrm{GREG}}(\beta)\} = \frac{1}{N^2}\sum_i \frac{1-\pi_i}{\pi_i}\{Y_i(1) - X_i^\top \beta\}^2.$$

HT is the case $\beta = 0$. At the other extreme, if the prediction were exact for every unit, every residual would be zero and the treated potential-outcome mean could be recovered from full-roster predictions without assignment noise. Between those extremes, the fixed-coefficient estimator reduces variance when the chosen coefficient leaves less design-weighted residual variation than the raw outcomes; an arbitrary or poorly transported coefficient need not help.

For control, replace Z_i/π_i by $(1-Z_i)/(1-\pi_i)$ and use a control-outcome prediction m_0. Subtracting the two prediction-plus-residual arm means estimates the ATE. This is the separate-arm version of the adjustment in Equation 14.

In practice, β is often fitted from the realized outcomes. Then the fixed-coefficient argument no longer establishes exact finite-sample unbiasedness. A coefficient fitted on historical data can be frozen before assignment; a coefficient fitted within the experiment needs the regression-adjustment qualifications discussed above. L03 uses the prediction-plus-IPW construction for both arms and names the resulting general form AIPW (Robins et al. 1994).

6.2 CUPED: choosing the control coefficient

The same online setting provides a natural scalar control variable: the outcome metric measured before the experiment. In this subsection, take X_i to be that single pre-period metric rather than a vector. **CUPED**, short for controlled-experiment using pre-experiment data, transforms the observed outcome to

$$Y_i^{\text{cuped}}(\theta) = Y_i^{\text{obs}} - \theta(X_i - \overline{X}).$$

Subtracting the same pretreatment quantity from $Y_i(1)$ and $Y_i(0)$ does not change unit i's treatment effect. Centering at $\overline{X}$ makes the transformed outcome retain the original finite-population mean within each treatment state. Taking the difference in arm means therefore gives, equivalently,

$$\hat{\tau}_{\text{CUPED}}(\theta) = \hat{\tau}_{\text{DM}} - \theta\hat{\tau}_X. \tag{17}$$

Here θ has the units of outcome per unit of the pre-period metric: it says how much outcome difference should be removed for one unit of chance covariate imbalance. For every θ fixed before assignment, this estimator remains design-unbiased because $\mathbb{E}_Z(\hat{\tau}_X) = 0$. The choice of θ controls precision. Expanding the variance of $\hat{\tau}_{\text{DM}} - \theta\hat{\tau}_X$ gives

$$\begin{aligned}\operatorname{Var}_Z\{\hat{\tau}_{\text{CUPED}}(\theta)\} = {} & \operatorname{Var}_Z(\hat{\tau}_{\text{DM}}) + \theta^2 \operatorname{Var}_Z(\hat{\tau}_X) \\ & - 2\theta \operatorname{Cov}_Z(\hat{\tau}_{\text{DM}}, \hat{\tau}_X).\end{aligned} \tag{18}$$

When $\operatorname{Var}_Z(\hat{\tau}_X) > 0$, this quadratic is minimized at

$$\theta^* = \frac{\operatorname{Cov}_Z(\hat{\tau}_{\text{DM}}, \hat{\tau}_X)}{\operatorname{Var}_Z(\hat{\tau}_X)}. \tag{19}$$

This is an oracle coefficient: its covariance with the treatment-effect estimator depends on the potential-outcome schedule and is not known before the experiment. At this value, the variance reduction relative to unadjusted difference in means is $\operatorname{Cov}_Z(\hat{\tau}_{\text{DM}}, \hat{\tau}_X)^2 / \operatorname{Var}_Z(\hat{\tau}_X)$. Thus θ should

be near zero when the pre-period measure does not predict experiment-period outcomes, and a poor coefficient can increase rather than decrease variance.

Under complete randomization, define the finite-population variance and arm-specific outcome–covariate covariances by

$$S_X^2 = \frac{1}{N-1}\sum_i (X_i - \overline{X})^2, \qquad S_{zX} = \frac{1}{N-1}\sum_i \{Y_i(z) - \overline{Y}(z)\}(X_i - \overline{X}).$$

The complete-randomization covariance identities are

$$\mathrm{Var}_Z(\hat{\tau}_X) = S_X^2\left(\frac{1}{N_1} + \frac{1}{N_0}\right),$$

and

$$\mathrm{Cov}_Z(\hat{\tau}_{\mathrm{DM}}, \hat{\tau}_X) = \frac{S_{1X}}{N_1} + \frac{S_{0X}}{N_0}.$$

The second expression has a plus sign because both $\hat{\tau}_{\mathrm{DM}}$ and $\hat{\tau}_X$ subtract their control-arm means. Substituting these identities into Equation 19 gives

$$\theta^* = \frac{S_{1X}/N_1 + S_{0X}/N_0}{S_X^2(1/N_1 + 1/N_0)} = \frac{N_0}{N}\frac{S_{1X}}{S_X^2} + \frac{N_1}{N}\frac{S_{0X}}{S_X^2}.$$

The oracle coefficient is therefore an allocation-weighted average of the two arm-specific slopes, with each slope weighted by the size of the opposite arm. Under equal allocation it is their simple average; if the two slopes coincide, their common value is optimal for any allocation. This calculation explains both why a pooled slope can work well and why treatment-specific slopes can improve on it.

The original CUPED recommendation is to use the same metric from a nearby pre-experiment period: it is often strongly correlated with the experiment-period metric while remaining unaffected by treatment (Deng et al. 2013). There are several practical ways to choose θ:

- **Estimate it before the experiment.** Use comparable historical experiments, a separate cohort, or relationships between earlier pretreatment windows, then freeze the coefficient before assignment. This preserves the exact fixed-coefficient design argument, but relies on stability across periods.
- **Estimate the oracle combination in the experiment.** Estimate a slope within each arm, say $\hat{\beta}_1$ and $\hat{\beta}_0$, and combine them as $(N_0/N)\hat{\beta}_1 + (N_1/N)\hat{\beta}_0$, mirroring the last expression for θ^*. In a balanced experiment this is the simple average. Regressing Y_i^{obs} on an intercept, Z_i, and $X_i - \overline{X}$ gives a familiar pooled common slope; including Z_i means that the slope is learned from within-arm outcome–covariate association rather than from the treatment/control mean difference. In a large balanced experiment it provides a convenient estimate of the same average when both arms inherit similar covariate dispersion; ordinary pooled regression need not reproduce the oracle weighting in an unbalanced experiment. Because the same outcomes choose the coefficient and estimate the effect, exact finite-sample unbiasedness is no longer automatic; inference should account for the fitted adjustment.

- **Allow treatment-specific slopes when warranted.** Include $Z_i(X_i - \overline{X})$, or fit the two arm-specific slopes directly. A single common θ can leave precision unused when the prognostic relationship differs by arm (Lin 2013).
- **Regularize or cross-fit richer adjustments.** With many baseline variables or a flexible predictor, assess predictive performance out of sample and use sample splitting or cross-fitting to limit overfitting. Cross-fitting does not by itself guarantee exact design unbiasedness, especially under fixed-total assignment. Only pretreatment variables belong in the adjustment.

7 Calibration: make the weighted arm reproduce its target

Return first to the completely randomized account experiment. Its treated-arm design weights are $d_i = 1/\pi_i = N/N_1$. They sum to N automatically, but if the treated users average 4.8 pre-period active days, their weighted covariate total is $N(4.8)$ rather than the known roster total $N(4.0)$. With unequal treatment probabilities, the same issue can affect both the total weight and the covariate total: design weights reproduce the full-roster targets in expectation over assignments, not necessarily in the realized assignment.

Calibration replaces the base weights by nearby weights that make selected weighted facts about the observed arm equal known full-roster targets. Its goal is not to make the treated arm resemble the roster in every respect. It enforces the particular comparisons encoded by the analyst's chosen features. In the scalar example, it chooses weights w_i satisfying

$$\sum_{i:Z_i=1} w_i = N, \qquad \sum_{i:Z_i=1} w_i X_i = N\overline{X}. \tag{20}$$

The first equation makes the weighted treated arm represent N accounts. The second makes its weighted pre-period total equal the known full-roster total. Under complete randomization the base weights already satisfy the first equation, but adjusting them to satisfy the second can disturb that total, so both constraints are imposed together.

For several covariates, first choose K scalar calibration features $h_1(X_i), \ldots, h_K(X_i)$. These can be individual coordinates of X_i, indicators for categories, or prespecified transformations and interactions. They express which roster summaries the weighted arm should match. Collect them with an intercept into the $(K+1)$-dimensional column vector

$$g_i = \begin{pmatrix} 1 \\ h_1(X_i) \\ \vdots \\ h_K(X_i) \end{pmatrix}, \qquad t_g = \sum_{i=1}^{N} g_i.$$

The leading 1 is an intercept coordinate; g_i is an ordinary vector, not a binomial coefficient. Because the pretreatment covariates are known for the full experimental roster, the target total t_g is known. The calibration equation is the vector equality

$$\sum_{i:Z_i=1} w_i g_i = t_g. \tag{21}$$

Written component by component, it requires

$$\sum_{i:Z_i=1} w_i = N, \qquad \sum_{i:Z_i=1} w_i h_k(X_i) = \sum_{i=1}^{N} h_k(X_i), \quad k = 1, \ldots, K.$$

No binary-covariate assumption is involved. For a binary feature, the equation matches a count; for a continuous feature, it matches a weighted sum, and therefore its mean once the intercept constraint holds.

The constraints usually admit many weight vectors: matching $K+1$ totals does not by itself say which unit should absorb each adjustment. Let $\mathbf{w}$ collect the treated-arm weights. Calibration resolves that ambiguity by starting from the design weights d_i and moving them as little as possible while hitting the targets:

$$\min_{\mathbf{w}} \sum_{i:Z_i=1} d_i \, \phi\left(\frac{w_i}{d_i}\right) \quad \text{subject to} \quad \sum_{i:Z_i=1} w_i g_i = t_g, \tag{22}$$

where the convex function ϕ is minimized at 1. Thus the objective favors w_i/d_i near one, while the constraints force exactly as much movement as is needed for balance. With a strictly convex distance and feasible constraints, the solution is unique. Quadratic distance, $\phi(r) = (r-1)^2/2$, gives affine adjustments

$$w_i = d_i(1 + \eta^\top g_i),$$

whereas relative-entropy distance, $\phi(r) = r \log r - r + 1$, gives the positive exponential tilt

$$w_i = d_i \exp(\eta^\top g_i).$$

In each case $\eta \in \mathbb{R}^{K+1}$ is a vector of Lagrange-multiplier parameters chosen so that Equation 21 holds. Entropy calibration becomes classical raking when the features are indicators for categorical margins (Deville and Särndal 1992; Deville et al. 1993).

The resulting treated-mean estimator is

$$\hat{\mu}_{1,\text{cal}} = \frac{1}{N} \sum_{i:Z_i=1} w_i Y_i^{\text{obs}}.$$

Its potential advantage follows directly from the balance equations. This does not require assuming a linear outcome model. For any chosen β, define the residuals algebraically by

$$Y_i(1) = g_i^\top \beta + e_i.$$

Then

$$\hat{\mu}_{1,\text{cal}} - \overline{Y}(1) = \frac{1}{N} \left\{ \sum_{i:Z_i=1} w_i e_i - \sum_{i=1}^{N} e_i \right\}. \tag{23}$$

To obtain Equation 23, substitute $Y_i(1) = g_i^\top \beta + e_i$ into both the weighted observed total and the full-roster total. The two $g_i^\top \beta$ contributions are equal because of Equation 21, so they cancel. The only remaining error is imbalance in the residuals.

This identity gives the calibration objective its statistical meaning. If the chosen features predict outcomes well, their exact balance removes a systematic source of assignment noise. If $Y_i(1) = g_i^\top \beta$ exactly for every unit, all residuals are zero and the calibrated treated mean equals $\overline{Y}(1)$ exactly. With weakly predictive or overly numerous features, the benefit can be small or outweighed by unstable weights.

When the calibration problem is feasible and the relevant weighted feature matrix has full rank, quadratic distance gives the calibrated estimator an exact GREG representation with a corresponding weighted least-squares coefficient. Other smooth calibration distances generally produce different finite-sample estimators, although they can share a first-order GREG approximation under regularity conditions (Deville and Särndal 1992). Unlike the generalized-difference construction with a coefficient fixed before assignment, calibrated weights depend on the realized assignment, so exact design unbiasedness is not automatic.

The control arm can be calibrated separately to the same roster targets, beginning from $d_i = 1/(1-\pi_i)$. Subtracting the calibrated arm means estimates the ATE. More constraints are not always better: they can force extreme weights and inflate the residual variance. With nonnegative weights and an intercept constraint, the target covariate mean must lie in the observed arm's convex hull. Signed quadratic weights can reach targets outside that hull only when the target remains in the affine span of the observed feature vectors.

Diagnostics should therefore include residual balance, weight dispersion, and sensitivity to the feature set. For positive weights, one useful summary is the Kish effective sample size

$$N_{\text{eff}} = \frac{(\sum_{i:Z_i=1} w_i)^2}{\sum_{i:Z_i=1} w_i^2}.$$

It equals the observed arm size under equal weights and decreases as a small number of units carry more of the weighted total. It is a warning signal about weight concentration, not by itself a complete variance calculation.

8 What to carry forward

Many estimators in this lecture share one architecture, either exactly or through a first-order approximation:

unbiased or identifying core + a correction with known or approximately known mean zero.

HT corrects unequal observation probabilities. Hájek stabilizes random total weight. Trotter–Tukey chooses how much to normalize. Regression, CUPED, GREG, and calibration remove predictable imbalance. Some corrections preserve exact unbiasedness only when their coefficients are fixed relative to assignment; fitted corrections generally require additional finite-sample or asymptotic

analysis. Their differences matter, but seeing the common control-variate structure makes the menu navigable.

9 Advanced material

9.1 Practical experimentation: errors larger than the standard error

Large samples do not rescue a broken experiment. The practical program begins with Kohavi et al. (2007); Crook et al. (2009) then documents recurring failures in online controlled experiments: incorrect variance and power calculations, misleading percent-change intervals, bot traffic, instrumentation failures, ramp-up effects, and treatment/control populations that are not actually comparable. These are failures of measurement and implementation before they are failures of asymptotic approximation.

The replication evidence in Kohavi et al. (2026) adds a selection lesson. Eight large-scale tests of previously reported experimentation "patterns" had median samples in the millions and small minimum detectable effects, yet only two produced significant movement in the expected direction and one moved significantly in the opposite direction. The discrepancy is consistent with underpowered discovery studies, selective reporting, and the **winner's curse**: conditional on being selected for a large estimate, a noisy estimate is likely exaggerated.

Practical discipline follows: establish sample-ratio and instrumentation checks, specify the primary metric and analysis before looking, ramp cautiously, report uncertainty in interpretable units, and replicate claims that will guide many future experiments.

9.2 The propensity-score paradox

Why estimate an assignment probability that the experimenter already knows? The efficiency-bound analysis of Hahn (1998) sets up the issue. In a semiparametric superpopulation setting, Hirano et al. (2003) show that inverse-probability estimators using an estimated propensity score can be more efficient than otherwise analogous estimators using the true propensity.

There is no information paradox. Common likelihood and series estimators of the propensity score satisfy empirical fitting equations that can also impose useful covariate-balance conditions. Those conditions create a projection or control-variate correction, much like Equation 13 and Equation 21; arbitrary propensity estimation need not impose the same balance. The estimated propensity is not better knowledge of the randomization device. It is a route to an estimator that exploits realized covariate balance. In a randomized experiment, it is clearer to describe the gain directly as covariate adjustment or calibration, while retaining the known design probabilities as the basis for identification and inference.

9.3 A short history of inverse-probability normalization

Horvitz and Thompson used inverse inclusion probabilities to estimate finite-population totals exactly unbiasedly (Horvitz and Thompson 1952). Ratio estimation and Hájek normalization can trade finite-sample bias for stability; whether that trade is favorable depends on the design and finite

population. Trotter and Tukey placed normalization on a continuum (Trotter and Tukey 1956), while Basu's survey-sampling arguments exposed how formally unbiased estimators can behave poorly when they ignore structural information (Basu 1971).

The same ratio appears in importance sampling: self-normalized importance sampling is stable and invariant to an unknown normalizing constant, but biased in finite samples (Owen 2013). Counterfactual learning rediscovered the same practical tradeoff when propensities are highly variable (Swaminathan and Joachims 2015). Adaptive normalization makes the amount of normalization estimable and connects its fixed point to regression control (Khan and Ugander 2023). The recurring problem is not domain-specific: random weights carry useful correction and avoidable noise at the same time.

9.4 Choosing estimators across families of experiments

Parikh et al. (2026) argue against choosing whichever estimator looks best inside a single realized trial. That practice turns estimator selection into another opportunity for overfitting. They instead use sample splitting across a **family** of randomized trials to compare estimators under a prespecified objective.

Their applications make the objective dependence concrete: weighted least squares performed best for inference-oriented criteria, while difference in means performed best for decision regret in the examined settings. The point is not a new universal winner. It is that "best estimator" is incomplete without a loss function, a relevant family of experiments, and a selection procedure separated from final evaluation.

10 Further reading

For the survey-sampling foundations, begin with Horvitz and Thompson (1952) and the exchange between Basu (1971) and Hájek (1971), then use Särndal et al. (1992) for the systematic development. The GREG and calibration line runs through Cassel et al. (1976), Deville and Särndal (1992), and Deville et al. (1993). For regression adjustment in experiments, reading Freedman (2008) together with Lin (2013) shows both the finite-sample concern and the modern interacted-regression resolution.

The normalization sequence from classical conditional Monte Carlo to adaptive IPW runs through Trotter and Tukey (1956) and Khan and Ugander (2023); Owen (2013) and Swaminathan and Joachims (2015) show the same self-normalization tradeoff in importance sampling and counterfactual learning. Deng et al. (2013) gives the original CUPED presentation in online experimentation.

For extensions beyond the main lecture path, read Hahn (1998) with Hirano et al. (2003) for the semiparametric efficiency logic behind estimated propensities. Kohavi et al. (2007), Crook et al. (2009), and Kohavi et al. (2026) connect estimator validity to experimentation practice and selective replication, while Parikh et al. (2026) treats estimator choice itself as a problem to be evaluated across a family of experiments.

Lecture brief: View the source `brief.md` · How these notes are made

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